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Performance evaluation and multi-objective optimization of a hybrid earth-air heat exchanger and building-integrated photovoltaic/thermal system with phase change material and exhaust air heat recovery

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ABSTRACT

The current study explores the feasibility of employing an integrated system comprising an earth-air heat exchanger (EAHE) unit and a photovoltaic/thermal (HPT) system equipped with a phase change material (PCM) to fulfill the heating, cooling, and electrical loads of a building. In this arrangement, ambient air is heated during cold months by passing it through the EAHE and PCM-based HPT systems, and during hot months, it is cooled by passing through the EAHE unit. Additionally, in warm months, the air exiting the building is utilized to lower the temperature of the PV panels. The electricity generated by the PV panels is harnessed to meet a portion or the entirety of the building's electrical demand. Following the identification of factors influencing the system performance, the multi-objectives genetic algorithm (MOGA) approach is employed to ascertain the optimal values of these parameters, aiming to concurrently maximize both the thermal and electrical output of the HPT-EAHE system. The findings indicated that the optimal HPT-EAHE system can fulfill 132693.3 kWh of heating demand for the building during cold months, address 26536.1 kWh of cooling needs during hot months, and generate 9242.2 kWh electrical power throughout the entire year.

1. Introduction

Buildings stand as silent giants in the battle against climate change, yet they're also some of the largest contributors to global warming due to their voracious appetite for energy [1]. Whether towering skyscrapers or humble homes, buildings devour energy for heating, cooling, lighting, and powering various devices, thereby releasing staggering amounts of greenhouse gases into the atmosphere [2]. The reliance on fossil fuels to meet these energy needs amplifies the crisis, as burning coal, oil, and gas emits carbon dioxide and other pollutants, intensifying the greenhouse effect and driving climate change. This incessant cycle of energy consumption not only accelerates global warming but also perpetuates environmental degradation, threatening ecosystems and human livelihoods worldwide. However, amidst this bleak scenario lies a beacon of hope: the transition to renewable energies [3]. Embracing solar, wind, hydroelectric, and other renewable sources offers a lifeline in the fight against climate change. Unlike their fossil fuel counterparts,

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renewable energies produce minimal greenhouse gas emissions and promise a sustainable future for the built environment [4].

Photovoltaic (PV) panels stand at the forefront of renewable energy solutions, playing a pivotal role in combating global warming by harnessing the sun's inexhaustible energy. These innovative devices convert sunlight directly into electricity, offering a clean and sustainable alternative to fossil fuels. However, one of the challenges faced by PV panels is the reduction in efficiency as they heat up [5]. To address this issue, hybrid photovoltaic/thermal (HPT) systems have emerged as a promising solution. These systems combine the benefits of PV and thermal technologies, allowing for simultaneous electricity generation and thermal energy capture [6]. By utilizing the heat absorbed by solar panels to produce hot water or air, HPT systems improve overall efficiency and maximize the utilization of solar energ [7]. This integration not only enhances the performance of solar panels but also increases the versatility and applicability of solar energy solutions, paving the way for a more sustainable and efficient future [8].

Earth-air heat exchanger (EAHE) systems represent an innovative and sustainable approach to heating, cooling, and ventilating buildings [9]. These systems utilize the relatively stable temperature of the ground, typically a few meters below the Earth's surface, to exchange heat with the air [10]. In colder months, the EAHE system pre-warms incoming air, while in warmer months, it cools the air before it enters the building [11]. This energy-efficient technology capitalizes on the Earth's thermal mass, providing a cost-effective and environmentally friendly solution for maintaining comfortable indoor temperatures. EAHE systems contribute to reduced reliance on traditional heating and cooling methods, fostering energy conservation and sustainable building practices [12]. Considering the notable characteristics of HPT and EAHE systems, along with their applicability in buildings, some researchers suggested the concept

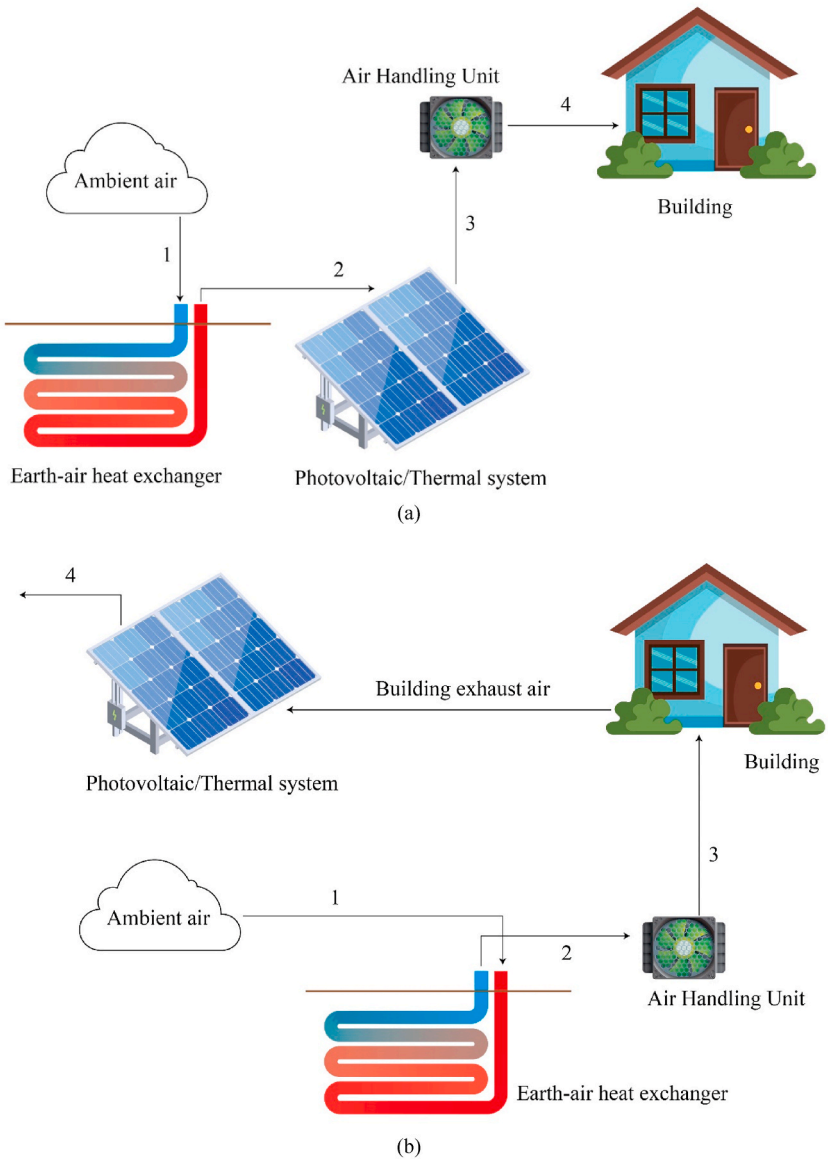


Fig. 1. Operational concepts of the HPT-EAHE unit: (a) heating mode and (b) cooling mode.

of utilizing both simultaneously to provide heating, cooling and electrical loads of buildings. Nayak and Tiwari [13] conducted a study evaluating the annual performance of an HPT-EAHE unit integrated with a greenhouse at IIT Delhi, India. Results highlight Jodhpur as the most favorable climatic condition, exhibiting the highest overall thermal energy, electrical energy savings, and exergy. Jakhar et al. [14] explored the thermal performance of an HPT-EAHE unit. Results indicated that the EAHE's heating capacity is notably enhanced with the addition of the PV/T air collector. Mahdavi et al. [15] conducted a theoretical examination of the energy and exergy efficiencies of a HPT-EAHE unit integrated with a greenhouse. The findings indicated that HPT unit did not possess notable potential for heating to elevate greenhouse air and plant temperatures. Yadav et al. [16] formulated a periodic thermal model for a semi-transparent HPT-EAHE unit integrated into a greenhouse. This model took into account design and climatic parameters. The results suggested that at a flow rate of 0.5 kg/s, the EAHE provides an extra thermal energy input to the HPT system, resulting in a rise of 4 °C in plant temperature and 5 °C in greenhouse air temperature. Jilani et al. [17] examined the output of a thin-film HPT-EAHE unit integrated with a quonset-type greenhouse. The findings reveal that the earth-air heat exchanger (EAHE) effectively provides heating within the enclosed space of the HPT system during non-sunshine hours, leading to reduced room air temperature fluctuations and improved thermal comfort. Shahsavar and Arici [18] tried to improve the performance of an HPT-EAHE system by adding heat pipes to the HPT unit. Two-objective optimization was employed to concurrently maximize the system's annual energy and exergy performance. Adding the heat pipes was associated with weakening the energy performance and strengthening the exergy performance of the system.

PCMs constitute a class of substances renowned for their exceptional ability to absorb and release thermal energy during phase transitions, such as melting or solidifying [19]. This unique property finds versatile applications, notably within the PV and HPT systems [20]. In the context of these systems, PCMs act as thermal energy storage solutions, addressing the intermittency of sunlight. During periods of peak solar exposure, when excess energy is generated, PCMs efficiently absorb and store this thermal energy. Subsequently, during periods of reduced sunlight or heightened energy demand, the PCM undergoes a phase transition, releasing stored thermal energy to supplement the PV system's power output. This strategic integration not only enhances the overall efficiency of PV and HPT systems but also contributes to energy resilience, ensuring a consistent and reliable power supply by effectively managing thermal fluctuations [20]. Numerous investigations have focused on enhancing the efficiency of HPT and EAHE systems through the utilization of PCM [21,22]. However, a literature review highlights the absence of studies on the performance of HPT-EAHE systems incorporating PCM. Consequently, the current numerical research is dedicated to addressing this research gap. The examined HPT-EAHE system has the capability to warm ambient air during cold season, cool it in hot season, and provide the necessary electricity for a building. Additionally, this system possesses the benefit of recovering heat from the air exiting the building. First, the parameters affecting the HPT-EAHE system performance are determined. Then, two-objective optimization is used to determine the value of these parameters to simultaneously maximize the thermal and electrical power outputs of the system. Finally, the hourly, monthly and yearly performance of the optimal HPT-EAHE system is investigated. All calculations have been done for the weather conditions of Kermanshah, Iran (34.33°N, 47.08°E). According to data from the Iranian Meteorological Organization, Kermanshah city receives solar radiation with an intensity of 7045 MJ/m^2, and it typically enjoys around 280 sunny days annually [23].

2. The operational concepts of the HPT-EAHE unit

Fig. 1 illustrates the operational principles of the HPT-EAHE system. The building's designated indoor temperature is set at 23 °C. If the HPT-EAHE device fails to achieve this temperature with ambient air before entering the building, the air handling unit takes on the

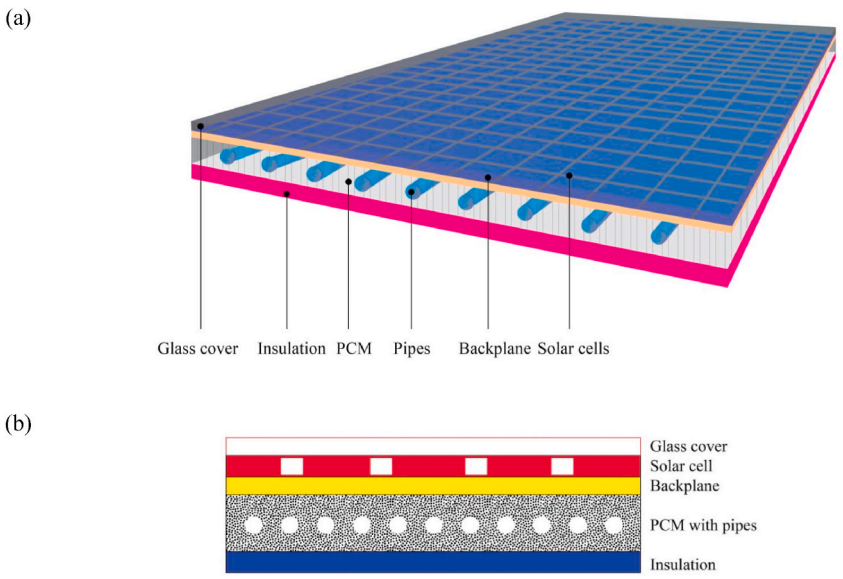


Fig. 2. (a) Illustrative and (b) cross-sectional perspectives of the PCM-HPT unit [24].

responsibility of raising the discharged air temperature from the HPT-EAHE device to meet the 23 °C requirement. When the ambient air temperature falls below 23 °C, a blower is employed to circulate the ambient air initially through the EAHE system and subsequently through the HPT system. This process elevates the air temperature by extracting heat from both the ground and the PV panel. Simultaneously, the temperature of the PV panels decreases, enhancing their efficiency. Conversely, when the ambient air temperature exceeds 23 °C, it is directed into the EAHE system to lower its temperature through heat transfer to the ground. In this scenario, the responsibility for cooling the PV panels rests on the air leaving the building, which possesses a lower temperature than the ambient air.

Fig. 2 illustrates the constituents of the PCM-HPT system. The setup comprises a glass cover, PV panel, aluminium plate, a PCM compartment, copper pipes enveloped by PCM, and insulation. The glass cover shields the solar cells from adverse weather conditions, such as rain and snow. Simultaneously, the aluminium plate facilitates the efficient transfer of excess heat from the PV panels to the PCM. Air circulates through the copper pipes, extracting surplus heat from the PV panels via the PCM and the pipe walls before expelling it from the system. Insulation is incorporated to impede the dissipation of heat stored in the PCM to the surrounding air. The incorporation of PCM in this layout offers two notable superiorities.

- Maintaining the temperature of PV panels within a specific range near ambient temperature, thereby enhancing their output.
- Retaining heat in PCMs even after sunset, enabling air heating during periods without sunlight.
- It is essential to highlight that seamlessly transitioning between the cooling and heating modes of the integrated system is facilitated by an integration of dampers and channels.

3. Features of numerical study

3.1. Governing equations

The procedural approach utilized for mathematically representing the performance of the HPT-EAHE unit involves individual modelling of the PCM-HPT and EAHE devices using the energy conservation and ε-NTU methods, respectively. Subsequently, these models are interconnected. The assumption is made that sunlight intensity and ambient temperature remain constant within a given hour but vary across different hours. One sample day is selected for each month, and the results obtained are extrapolated for the entire month. The subsequent part of this section delineates the distinct modelling methods employed for each system.

3.1.1. PCM-HPT system

To analyse the functioning of the PCM-HPT unit, the conservation of energy is established for each element, and then these equations are resolved to specify the temperature of each component. These equations operate under the assumptions that the received radiation is not reflected, the temperatures of different system components remain spatially constant, except for the flowing air, and the heat capacity of components other than the PCM layer is negligible [24].

The conservation of energy for various system layer is as follows [25].

- Glass cover:

$$\alpha_g I = h_{ge}\left(T_g - T_e\right) + U_{gc}\left(T_g - T_c\right) \tag{1}$$

- PV panel:

$$F\alpha_c\tau_g I + U_{gc}\left(T_g - T_c\right) = 0.165F\alpha_c\tau_g I[1 - 0.0045(T_c - 298)] + U_{cb}(T_c - T_b) \tag{2}$$

- Back plate:

$$F\alpha_b\tau_c\tau_g I + (1 - F)\alpha_b\tau_g I + U_{cb}(T_c - T_b) = U_{bp}\left(T_b - T_p\right) \tag{3}$$

- PCM compartment:

The surplus heat generated by the PV panels is conveyed to the PCM through conduction and natural convection processes. When the Rayleigh number is below 1000, the dominance of the conduction mechanism surpasses that of natural convection [26].

$$\delta Q_p = WL\left[U_{bp}\left(T_b - T_p\right) - h_{pw}\left(T_p - T_w\right) - U_{pi}\left(T_p - T_i\right)\right]\delta t$$

$$\begin{cases} \text{If } Q_p < Q_{sol}, \\ T_p = T_{mp} - \dfrac{Q_{sol} - Q_p}{c_{P\ in\ solid\ state}}, \lambda_p = \lambda_{PCM\ in\ solid\ state} \end{cases}$$

$$\begin{cases} \text{If } Q_{sol} < Q_p < Q_{liq}, \\ T_p = T_{mp}, \lambda_p = \lambda_{PCM\ in\ solid\ state}\dfrac{Q_{liq} - Q_p}{Q_{liq} - Q_{sol}} + \lambda_{PCM\ in\ liquid\ state}\dfrac{Q_p - Q_{sol}}{Q_{liq} - Q_{sol}} \end{cases} \tag{4}$$

$$\begin{cases} \text{If } Q_{liq} < Q_p, \\ T_p = T_{mp} - \dfrac{Q_p - Q_{liq}}{c_{P\ in\ liquid\ state}}, \lambda_p = \lambda_{PCM\ in\ liquid\ state} \end{cases}$$

- Air flow:

$$U_{pa}\left(T_p - T_a\right)\pi D dx = c_a \dot{m}_a dT_a \tag{5}$$

- Insulation layer:

$$U_{pi}\left(T_p - T_i\right) = h_{ie}\left(T_i - T_e\right) \tag{6}$$

where, α = absorption factor, I = intensity of sunlight, h = heat convection coefficient, F = packing factor, U = overall heat transfer coefficient, T = temperature, τ = transmissivity, Q_p = accumulated thermal energy influx into the PCM, Q_{sol} = enthalpy of solid-solid phase transition, Q_{liq} = enthalpy of solid-liquid phase transition, W = HPT width, L = HPT length, D = pipe diameter, c = heat capacity and $\dot{m}$ = mass flux. Moreover, the subscripts a, c, b, g, e, mp, p, and i respectively, signify the air flow, PV panel, back plate, glass cover, ambient air, melting point, PCM, and insulation.

The determination of U values in Eqs. (1)–(6) is achieved through the following calculations [25–30]:

$$U_{gc} = \left(\frac{\delta_g}{\lambda_g} + \frac{\delta_c}{\lambda_c}\right)^{-1} \tag{7}$$

$$U_{cb} = \left(\frac{\delta_c}{\lambda_c} + \frac{\delta_b}{\lambda_b}\right)^{-1} \tag{8}$$

$$U_{bp} = \left(\frac{\delta_b}{\lambda_b} + \frac{\delta_p}{\lambda_p}\right)^{-1} \tag{9}$$

$$U_{pa} = \left(\frac{1}{h_a} + \frac{\delta_t}{\lambda_t} + \frac{\delta_p}{\lambda_p}\right)^{-1} \tag{10}$$

$$U_{pi} = \left(\frac{\delta_p}{\lambda_p} + \frac{\delta_i}{\lambda_i}\right)^{-1} \tag{11}$$

where

$$h_a = \frac{\lambda_a}{D}\left\{0.0182\left(\frac{4\dot{m}_a}{\pi D \mu_a}\right)^{0.8}\left(\frac{\mu_a c_a}{\lambda_a}\right)^{0.4}\left[1 + j\frac{D}{L}\right]\right\} \tag{12}$$

$$j = 14.3\log\left(\frac{L}{D}\right) - 7.9 \quad \text{for} \quad 0 < \frac{L}{D} \le 60 = 17.5 \quad \text{for} \quad \frac{L}{D} > 60 \tag{13}$$

$$h_{ge} = \left(\frac{\delta_g}{\lambda_g} + \frac{1}{5.7 + 3.8 v_w}\right)^{-1} \tag{14}$$

$$h_{ie} = \left(\frac{\delta_i}{\lambda_i} + \frac{1}{5.7 + 3.8 v_w}\right)^{-1} \tag{15}$$

In which, δ = thickness and λ = thermal conductivity, μ_a = air viscosity, v_w = wind speed and the subscript t stands for the tube.

The air temperature at the outlet and the mean temperature of the air leaving the HPT unit can be calculated in the following manner [25]:

$$T_a(L) = T_{a,in}.e^{-\pi DU_{pa}L/c_a\dot{m}_a} + T_p\left(1 - e^{-\pi DU_{pa}L/c_a\dot{m}_aL}\right) \tag{16}$$

$$\overline{T}_a = \frac{1}{L}\int_0^L T_a(x) \bullet dx = T_{a,in}\left\{\frac{1 - e^{-\pi DU_{pa}L/c_a\dot{m}_a}}{\pi DU_{pa}L/c_a\dot{m}_a}\right\} + T_p\left\{1 - \frac{1 - e^{-\pi DU_{pa}L/c_a\dot{m}_a}}{\pi DU_{pa}L/c_a\dot{m}_a}\right\} \tag{17}$$

Here, $T_{a,in}$ = Intake air temperature.

3.1.2. EABE unit

Within the EAHE, the exchange of heat occurs between the air flowing through the pipes in the ground through convection. Simultaneously, there is a conduction-based transfer of heat between the pipe walls and the surrounding soil. The effectiveness of the EAHE system for two cases $Re_{EAHE} < 2300$ and $2300 \le Re_{EAHE} < 5 \times 10^6$ can be calculated using Eqs. (18) and (19), respectively [26,27]:

$$\varepsilon = 1 - exp\left\{-3.66\frac{\pi\lambda_a L_{EAHE}}{\dot{m}_a c_a}\right\} \tag{18}$$

$$\varepsilon = 1 - exp\left\{-\frac{\pi\lambda_a L_{EAHE}}{\dot{m}_a c_a}\left[\frac{\left(\frac{[1.82\ \ln(Re_{EAHE})-1.64]^{-2}}{8}\right)(Re_{EAHE} - 1000)Pr}{1 + 12.7\sqrt{\frac{[1.82\ \ln(Re_{EAHE})-1.64]^{-2}}{8}}\left(Pr^{\frac{2}{3}} - 1\right)}\right]\right\} \tag{19}$$

where L_{EAHE} = EAHE channel length, Pr = Prandtl number = $\frac{\mu_a c_a}{\lambda_a}$, Re_{EAHE} = Reynolds number of EAHE unit = $\frac{4\dot{m}_a}{\pi\mu_a D_{i,EAHE}}$, and $D_{i,EAHE}$ = inner diameter of EAHE channel.

Effectiveness is the measure of actual heat transfer compared to the maximum potential heat transfer [26]:

$$\varepsilon = \frac{T_{out,EAHE} - T_{in,EAHE}}{T_{soil} - T_{in,EAHE}} \tag{20}$$

where $T_{in,EAHE}$ denotes the incoming air temperature, $T_{out,EAHE}$ signifies the outgoing air temperature, and T_{soil} represents the soil temperature. The unaltered temperature of the Earth refers to the average temperature that remains relatively stable at a depth of 1.5–2 m throughout the year. In the context of Kermanshah, this constant value is 295.3 K and can be considered as the yearly average outdoor air temperature [28].

Based on the given definition, the EAHE system's effectiveness can be computed using either Eq. (18) or (19). Subsequently, the air's exiting temperature from the system can be ascertained in the following manner:

$$T_{out,EAHE} = T_{in,EAHE} + \varepsilon\left(T_{soil} - T_{in,EAHE}\right) \tag{21}$$

3.2. Assessment of performance characteristics

The air circulating through the HPT-EAHE system obtains heat from two sources: the PV panels during daylight hours when sunlight is accessible and from the soil. The rate of useful heat received by the air from the HPT-EAHE system can be computed as follows:

$$\dot{Q} = \underbrace{\dot{m}_a c_a\left(T_{out,EAHE} - T_{in,EAHE}\right)}_{EAHE\ system} + \underbrace{\dot{m}_a c_a\left(T_{out,HPT} - T_{in,HPT}\right)}_{HPT\ system} \tag{22}$$

In this study, the temperature of the air entering the EAHE system is equivalent to the ambient air temperature (i.e., $T_{in,EAHE} = T_e$), while the temperature of the air entering the HPT system matches the temperature of the air exiting the EAHE system (i.e., $T_{in,HPT} = T_{out,EAHE}$).

In the HPT-EAHE system, electricity is generated by PV panels, and a portion of this generated electric power is utilized to meet the energy needs of the fan responsible for circulating air within the system. Consequently, the calculation for the effective electrical power generated by the combined system is as follows [29,30]:

$$\dot{E} = 0.165\alpha_c IWL\left[1 - 0.00451\left(T_{pv} - 298\right)\right] - 2\frac{\dot{m}_a^3}{\rho_a^2}\left\{\frac{8\left(k_{c,HPT} + 1\right)}{\pi^2 D^4} + \frac{0.12807L}{D^5 Re_{HPT}^{0.25}} + \frac{8\left(k_{c,EAHE} + 1\right)}{\pi^2 D_{i,EAHE}^4} + \frac{0.12807L_{EAHE}}{D_{i,EAHE}^5 Re_{HPT}^{0.25}}\right\} \tag{23}$$

where $k_{c,BIPVT}$ = inlet loss coefficient of HPT system, $k_{c,EAHE}$ = inlet loss coefficient of EAHE system, and Re_{HPT} = Reynolds number of HPT unit = $\frac{4\dot{m}_a}{\pi\mu_a D}$.

3.3. Verification information

The HPT-EAHE unit is mathematically modelled using the approach outlined in section 3.1, employing a custom MATLAB code. Before showcasing the outcomes, it is crucial to validate the accuracy of this code. To accomplish this, the code's results for the HPT system were cross-referenced with the experimental data provided by Brown et al. [31] for the back temperature of a PV panel in a PCM-HPT unit. The PCM in reference [31] consisted of a mixture of Palmitic Acid and Caprylic Acid with a mass ratio of 3/1, and its melting temperature was 22.8 °C. The comparison of the results between the two researches at various times throughout the day, as can be seen in Table 1, demonstrated the satisfactory precision of the code, with the highest disparity between the results of the two researches being 5.39 %.

The verification process for the mathematical model of the EAHE system involves comparing the outlet air temperature obtained in this study with the experimental data from Bansal et al. [32], conducted under varying flow velocities. The EAHE system under examination is an actual installation located in Ajmer (Western India), with observations conducted on March 12, 2009, and replicated on April 08, 2009. Table 2 illustrates the results of this comparative analysis. Notably, the variation between simulated and experimental results falls within the acceptable range of 5.48 %–8 % of the experimental outcomes.

4. Genetic algorithm based multi-objective optimization

Multi-objective optimization using GA (MOGA) is an advanced approach that addresses complex problems involving conflicting and diverse objectives. In this context, MOGA extends the traditional GA framework to efficiently explore and navigate trade-offs among multiple conflicting criteria. Unlike single-objective optimization, where a singular optimal solution is sought, MOGA aims to generate a set of diverse solutions, known as the Pareto front, which represents the optimal compromise solutions for various combinations of objectives [33]. The inherent ability of GA to maintain a diverse population, combined with specialized techniques such as Pareto dominance, crowding distance, and elitism, enables MOGA to identify a range of solutions that capture the trade-off relationships between conflicting objectives [33]. This makes MOGA particularly useful in decision-making scenarios where decision-makers need to evaluate and choose from a set of solutions, each offering a unique balance of trade-offs. The application of MOGA spans various fields, including engineering design, finance, resource allocation, and project management, where decision-makers grapple with the challenge of optimizing multiple, often conflicting, goals simultaneously.

In the present study, MOGA is employed to concurrently enhance both the energy and exergy performances of the HPT-EAHE unit. Examining the relationships presented in section 3.1 for modeling the HPT-EAHE unit reveals that assuming constant thermophysical and optical properties, the parameters affecting the system's performance are L, W, D, $\dot{m}_a$, $D_{i,EAHE}$, and L_{EAHE}. Therefore, these parameters with a change range of $L = 1$–10 m, $W = 0.5$–5 m, $D = 0.1$–0.5 m, $\dot{m}_a = 0.057$–5.7 kg/s, $D_{i,EAHE} = 0.1$–0.5 m, and $L_{EAHE} = 20$–100 m are considered as optimization decision parameters. The objective functions of the optimization involve the annual average values of η_{en} and η_{ex}. It should be noted that the intervals considered for the decision-making variables are based on the values that have been used in the studies in the literature.

5. Results and discussion

This section initiates by scrutinizing the impact of alterations in the decision-making variables on the optimization objective functions. When examining the impact of each decision-making variable, the rest of the variables are considered constant ($L = 5.0$ m, $W = 3.0$ m, $D = 0.3$ m, $\dot{m}_a = 0.1$ kg/s, $L_{EAHE} = 60$ m and $D_{i,EAHE} = 0.3$ m). Table 3 presents the fixed parameters included in the present study and their values.

Fig. 3 depicts the variations in $\dot{Q}$ and $\dot{E}$ relative to changes in L and W. Both the $\dot{Q}$ and $\dot{E}$ exhibit an upward trend as L and W are enhanced. For example, when the L increases from 1 m to 10 m in $W = 5.0$ m, there is a notable rise in $\dot{E}$ by 902.42 % and $\dot{Q}$ by 8.54 %. Moreover, at $L = 10$ m, expanding the W from 0.5 m to 5.0 m results in a respective increase of 866.16 % and 4.24 % in the $\dot{E}$ and $\dot{Q}$. By increasing the L, the surface area of the PV panel elevates, which causes, on the one hand, more solar radiation is absorbed by the PV panel and the electricity produced by the system increases, and on the other hand, more heat is transferred to the PCM and, as a result, air flow. In addition, the rise of L is associated with the increase in the volume of PCM in the system, which increases the heat transfer from the PV panel to the air flow and, as a result, increases the $\dot{Q}$ and $\dot{E}$. The adverse effect of increasing the L on the system performance is also an increase in the pressure drop of the air flow in the HPT system tubes, which leads to an increase in the power consumption of the fan to circulate the air in the system and, as a result, a decrease in the $\dot{E}$. The results presented in Fig. 3 reveal that

Table 1
Verification outlines for the HPT unit.

Time	The PV panel temperature (°C)		
	Current study	Ref. [31]	Difference (%)
10:00	34.41	35.91	4.18
12:00	51.72	54.29	4.73
14:00	52.11	54.91	5.10
16:00	46.42	48.92	5.11
18:00	30.21	31.94	5.42
20:00	25.52	26.51	3.73

Table 2
Verification outlines for the EAHE unit.

	Air velocity = 2 m/s			Air velocity = 4 m/s		
	Ref. [32]	Current study	Difference (%)	Ref. [32]	Current study	Difference (%)
Inlet temp.	43.7	43.7	0.00	43.1	43.1	0.00
Outlet temp.	31	29.3	5.48	32.5	29.9	8.00

Table 3
Parameters held constant in this work [28–30].

Parameter	Value	Parameter	Value	Parameter	Value
λ_g (W/m.K)	1	$c_{P\ in\ solid\ state}$ (kJ/kg.K)	2.90	τ_g	0.91
λ_c (W/m.K)	145	$c_{P\ in\ liquid\ state}$ (kJ/kg.K)	2.10	δ_g (m)	0.005
λ_b (W/m.K)	0.36	τ_c	0.09	δ_c (m)	0.0003
λ_a (W/m.K)	0.026	F	0.83	δ_b (m)	0.0001
$\lambda_{PCM\ in\ solid\ state}$ (W/m.K)	0.24	v_w	2 m/s	δ_p (m)	0.05
$\lambda_{PCM\ in\ liquid\ state}$ (W/m.K)	0.15	α_g	0.05	δ_t (m)	0.0008
λ_t	383	α_c	0.90	$\delta_i(m)$	0.05
λ_i	0.034	α_b	0.50	$k_{c,HPT}$	0.5
ρ_a (kg/m^3)	1.2	μ_a (Pa.s)	0.0000151	$k_{c,EAHE}$	1.5

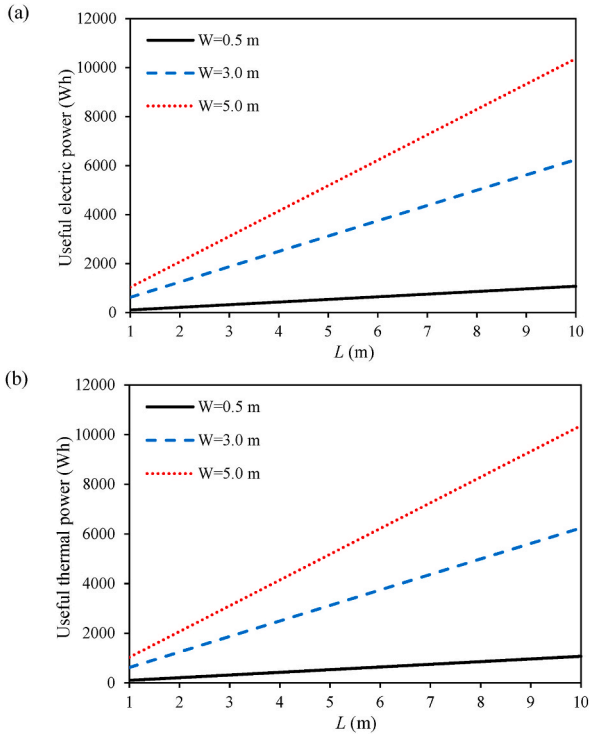


Fig. 3. Impact of L on the (a) $\dot{Q}$ and (b) $\dot{E}$ at various W s.

the increasing effect of L on the system performance overcomes its decreasing effect and as a result, the $\dot{Q}$ and $\dot{E}$ augment with increasing L. Expanding the W results in a rise in both the $\dot{Q}$ and $\dot{E}$ by enlarging the surface area of the PV panel, increasing the volume of PCM, and augmenting the quantity of airflow tubes.

Fig. 4 displays the variations in $\dot{Q}$ and $\dot{E}$ versus $\dot{m}_a$ and D. It is clear that as the $\dot{m}_a$ grows, the $\dot{E}$ initially rises and then declines, while the $\dot{Q}$ increases continuously. For instance, at $D = 0.5$ m, there's a 0.63 % increase in the $\dot{E}$ when the $\dot{m}_a$ augments from 0.0114 kg/s to 0.114 kg/s, whereas an increase in the $\dot{m}_a$ from 0.114 kg/s to 0.57 kg/s leads to a 17.04 % decrease in the $\dot{E}$. Moreover, boosting the $\dot{m}_a$ from 0.0114 kg/s to 0.57 kg/s results in a substantial 3669.63 % increase in the $\dot{Q}$. Furthermore, at $\dot{m}_a = 0.57$ kg/s, expanding the D from 0.1 m to 0.5 m is linked to a respective decrease of 4.77 % in the $\dot{E}$ and 2.89 % in the $\dot{Q}$. The augmentation in $\dot{m}_a$ contributes to an elevation in the convection heat transfer coefficient, leading to a subsequent increase in the rate of heat transfer to the airflow from

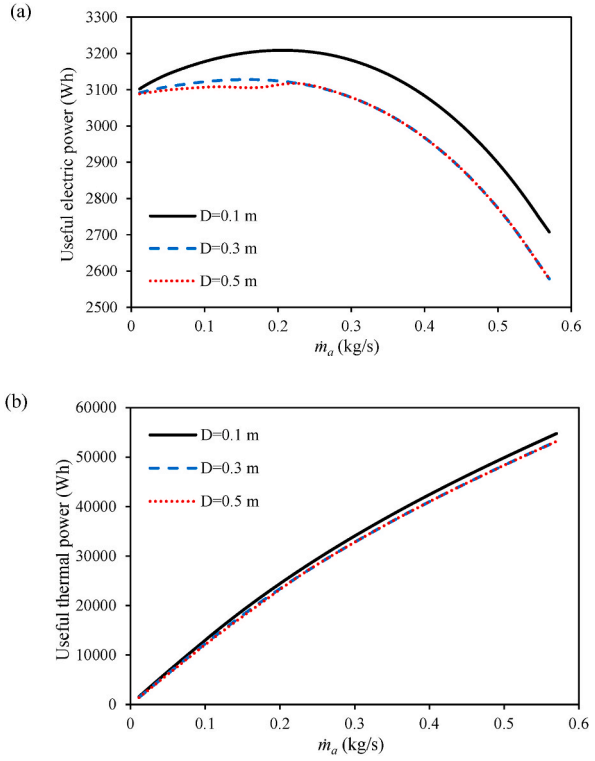


Fig. 4. Impact of $\dot{m}_a$ on the (a) $\dot{Q}$ and (b) $\dot{E}$ at various D s.

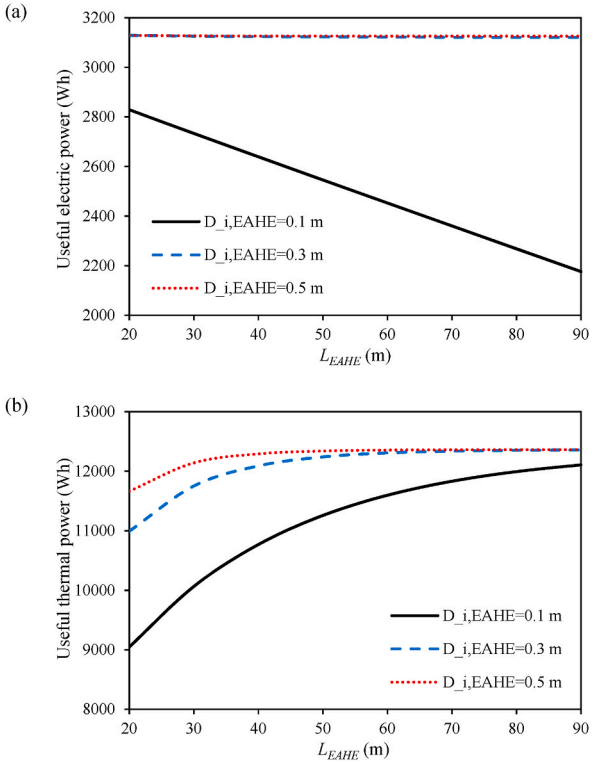


Fig. 5. Impact of L_{EAHE} on the (a) $\dot{Q}$ and (b) $\dot{E}$ at various $D_{i,EAHE}$ s.

both the HPT and EAHE systems. Consequently, there is an augmentation in both the $\dot{Q}$ and $\dot{E}$. Moreover, the rise in $\dot{m}_a$ correlates with a notable elevation in the pressure drop of the airflow within both the HPT and EAHE systems. This surge in pressure drop substantially raises the power consumption of the fans, leading to a consequent reduction in the $\dot{E}$. The examination of the findings indicates that up to a specific $\dot{m}_a$, the positive impact of the $\dot{m}_a$ on the $\dot{E}$ surpasses its negative influence, whereas at elevated $\dot{m}_a$ s, the reverse of this trend becomes apparent. Additionally, Fig. 4 illustrates that as the D increases, both the $\dot{Q}$ and $\dot{E}$ diminish, although the reduction is negligible for D s exceeding 0.3 m. An augmentation in the D, at a consistent $\dot{m}_a$, results in a reduction in the air velocity, leading to a decline in the rate of heat transferred to the air flow in the HPT system. Simultaneously, it induces an elevation in the PV panel temperature, consequently reducing the $\dot{E}$. Conversely, a reduction in the D is linked to a decrease in the pressure drop of air flow within the EAHE and HPT systems. Consequently, this results in lower power consumption by the fans, leading to a rise in the $\dot{E}$. The findings presented in Fig. 4 indicate that, up to a D of 0.3 m, the reducing impact of D on the $\dot{Q}$ and $\dot{E}$ surpasses its increasing effect. However, in larger diameters, the positive and negative effects balance each other out.

Fig. 5 shows the variations in $\dot{Q}$ and $\dot{E}$ in terms of L_{EAHE} and $D_{i,EAHE}$. As the L_{EAHE} increases, the $\dot{Q}$ increases and the $\dot{E}$ decreases. For example, at $D_{i,EAHE} = 0.1$ m, expanding the L_{EAHE} from 20 m to 90 m results in a 23.07 % decrease in the $\dot{E}$ and a 33.82 % rise in the $\dot{Q}$. Additionally, at $L_{EAHE} = 90$ m, enlarging the $D_{i,EAHE}$ from 0.1 m to 0.5 m causes a 43.63 % increase in the $\dot{E}$ and a 2.11 % increase in the $\dot{Q}$. It is evident that the rate of change of both parameters is insignificant for $D_{i,EAHE}$ s larger than 0.3 m. With the increase in L_{EAHE}, the rate of heat transferred to the flowing air in the EAHE system and the pressure drop of the air flow in the EAHE system increase, the former causes an increase in the $\dot{Q}$ and the latter causes a decrease in the $\dot{E}$. By increasing the $D_{i,EAHE}$ at a constant $\dot{m}_a$, the air flow speed decreases, which reduces the effect of the increase in L_{EAHE} on the $\dot{Q}$ and $\dot{E}$. In Fig. 5, it can also be seen that the increase in the $D_{i,EAHE}$ is accompanied by an increase in the $\dot{Q}$ and $\dot{E}$, the former is caused by a decrease in the speed of the fluid flow, and the latter is caused by a decrease in the flow pressure drop, and as a result, a decrease in the power consumption of the fan to flow air in the EAHE system.

The results presented so far indicated that the L, W, D, $\dot{m}_a$, L_{EAHE}, and $D_{i,EAHE}$ have a significant effect on the $\dot{Q}$ and $\dot{E}$. For this reason, these parameters are considered as the decision-making variables and the MOGA method is used to determine the value of these parameters, which leads to the simultaneous maximization of $\dot{Q}$ and $\dot{E}$. The Pareto curve resulting from the two-objective optimization is presented in Fig. 6. This curve includes 70 points, each of which can be considered an optimal point. The specifications of all these points are presented in Appendix A. Examining the value obtained for the optimization objective functions of these points shows that the maximum amount of $\dot{Q}$ and $\dot{E}$ is equal to 159211.3 Wh and 10826.76 Wh, respectively. None of the points in the Pareto chart have the highest amount of $\dot{Q}$ and $\dot{E}$ at the same time, and as a result, the point with the highest $\dot{Q}$ and $\dot{E}$ can be considered an ideal point. In the current research, the final optimal point is considered as the point that has the smallest spatial distance from the ideal point (L = 9.999 m, $W = 5.0$ m, $D = 0.1$ m, $\dot{m}_a = 1.73$ kg/s, $D_{i,EAHE} = 0.5$ m, $L_{EAHE} = 91.147$ m, $\dot{Q} = 159229.4$ kWh and $\dot{E} = 9242.2$ kWh).

The rest of this section is devoted to the investigation of the optimal HPT-EAHE system performance. The HPT-EAHE system offers a notable advantage by not only supplying the necessary hot and cold air for the building during winter and summer, respectively, but also facilitating the cooling of PV panels. Fig. 7 displays the system's success in accomplishing this on January 15 and August 15, representing cold and hot seasons. It is important to mention that the results presented pertain to hours with available sunlight. Additionally, Fig. 7 includes results for the non-cooling mode, calculated using the following formula [34]:

$$T_{PV-no\ cooling} = T_e + \frac{38}{800} I \tag{24}$$

The amount of PV panel cooling on January 15 and August 15 is 1.4–13.0 K and 11–35.6 K, respectively. The highest amount of cooling in both days belongs to 13:00, which shows that the HPT-EAHE system has been able to provide significant cooling in the hour when the intensity of radiation is significant.

Fig. 8 displays the hourly fluctuations in air temperature released from the HPT-EAHE system on both January 15 and August 15.

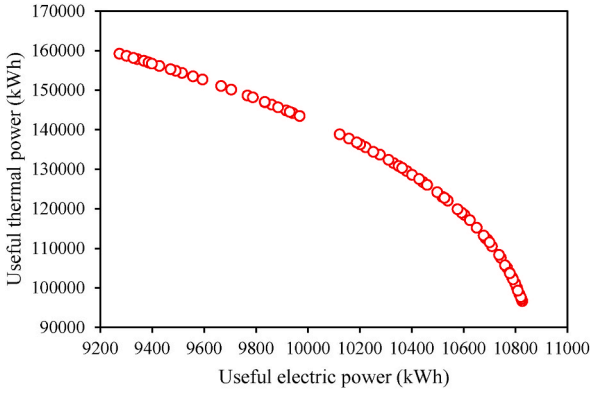


Fig. 6. Impact of L_{EAHE} on the $\dot{Q}$ and $\dot{E}$ at various $D_{i,EAHE}$ s.

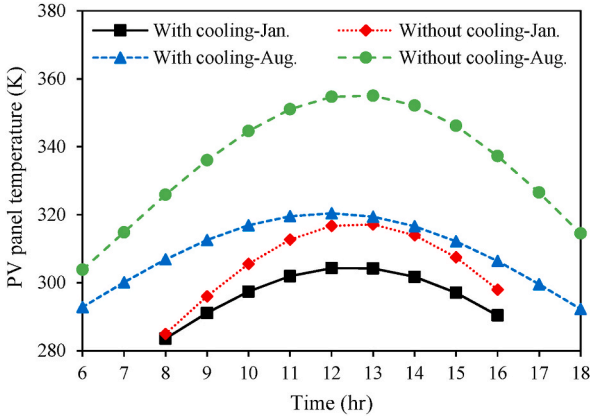


Fig. 7. PV panel temperature in the modes with and without cooling on January 15 and August 15.

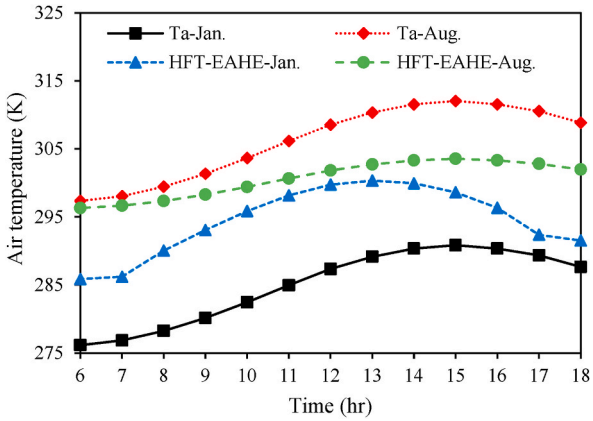


Fig. 8. PV panel temperature in the modes with and without cooling on January 15 and August 15.

To explain the success rate of the system in increasing/decreasing the outside air temperature on January 15/August 15, the ambient air temperature on these days is also displayed in Fig. 8. On January 15, the minimum and maximum heating levels, reaching 3.0 K and 13.2 K, respectively, occur at 11:00 and 17:00. Moreover, on August 15, the EAHE system is able to cool the ambient air by 0.9–8.5 K.

Fig. 9 illustrates the electric power generated by the optimal HPT-EAHE system and the contribution of both the HPT and EAHE systems throughout various months. Notably, the EAHE system consistently exhibits negative electrical power throughout the year, indicating its role as an electricity consumer rather than a producer. This is attributed to the energy required by the EAHE system's fan for air circulation. In contrast, the HPT system generates electric power ranging from 548.1 to 1312.4 kWh across different months, with July and December representing its peak and nadir values, respectively. Moreover, the electric power output of the optimal HPT-EAHE system falls within the 367.6–1131.9 kWh range, with December and July representing the minimum and maximum values,

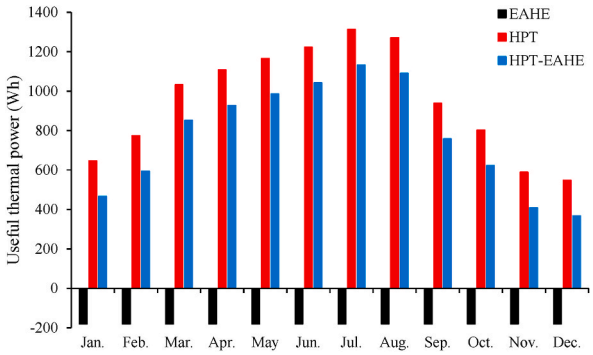


Fig. 9. The electrical performance of HPT, EAHE and HPT-EAHE systems throughout various months of the year.

respectively. Clearly depicted in Fig. 9 is the notable disparity in electricity generation between the hotter and colder months. This discrepancy is attributed to the increased intensity of solar radiation during the warmer periods of the year.

Fig. 10 illustrates the heat transfer rate to the air as it moves through the HPT, EAHE and HPT-EAHE systems during various months of a year. As previously mentioned, the HPT system does not contribute to cooling the ambient air during the warmer months. Consequently, during these periods, the useful thermal power generated by the EAHE and HPT-EAHE systems is identical. Based on the findings, the EAHE system releases its maximum (8474.9 kWh) and minimum (3042.7 kWh) thermal power to the ambient air in October and January, respectively. In contrast, the HPT and HPT-EAHE systems exhibit this pattern in November and March, respectively. The highest useful thermal power output of the HPT and HPT-EAHE systems is 20845.0 kWh and 25295.7 kWh, respectively, while the lowest value of this parameter for the mentioned systems is 14106.0 kWh and 17701.4 kWh, respectively. During January, February, March, October, November, and December, the contribution of the EAHE system to the beneficial thermal power output of the HPT-EAHE system is 34.7 %, 31.2 %, 17.6 %, 14.1 %, 20.3 %, and 32.1 %, respectively. Consequently, it appears that the HPT system has a more impactful role in heating mode compared to the EAHE system. The findings regarding the system's performance in the warm season of the year also indicate that in April, May, June, July, August, and September, the HPT-EAHE system can meet the cooling demands of the building at levels 3092.7 kWh, 3258.7 kWh,4209.2 kWh, 6738.9 kWh, 5434.8 kWh, and 3801.8 kWh, respectively.

It should be noted that the optimal HPT-EAHE system is capable of supplying 26536.1 kWh of cooling load, 132693.3 kWh of heating load, and 9242.2 kWh of electrical load of a building per year.

Ultimately, to examine how PCM influences the performance of the HPT-EAHE system, the useful thermal and electrical power output of the optimal HPT/PCM-EAHE system is contrasted with the results of optimal HPT-EAHE system analyzed by Li et al. [35]. It's worth mentioning that the decision variables chosen and their respective ranges of variation in Ref. [35] closely resemble those used in the current study. Based on the findings, incorporating PCM into the HPT-EAHE system has caused a 234 % increase in the total annual useful thermal power output of the system and a 0.2 % decrease in the total annual useful electric power output of the system. This significant improvement in the thermal performance of the system can be attributed firstly to the significant efficiency of PCM in the thermal management of PV panels and secondly to the ability of PCM to store thermal energy and transfer it to the air passing through the system even during the hours when sunlight It is not available. Weakening of the electrical performance of the HPT/PCM-EAHE system can also be explained by the fact that in the HPT-EAGHE system, the air passes directly through the channel located under the PV module, and as a result, its pressure drop is much lower than the HPT/PCM-EAHE system, where the air flows in a number of parallel pipes located inside the PCM. This makes the power consumption of the fan in the HPT-EAHE system much less than the HPT/PCM-EAHE system. Of course, a significant part of this difference is compensated due to the more electricity produced by the PV module in the HPT/PCM-EAHE system, but the results reveal that the electrical performance of the HPT-EAHE system is marginally better than the HPT/PCM-EAHE system.

6. Conclusion

The present investigation delves into the viability of implementing a PCM-based HPT/EAHE system to meet the heating, cooling, and electrical loads of a building located in Kermanshah, Iran. After identifying factors influencing system performance, the MOGA approach is applied to determine the optimal parameter values, aiming to simultaneously maximize both the thermal and electrical outputs of the HPT-EAHE system. The most important findings from the performed investigations are as follows.

- $\dot{Q}$ augments by increasing the L, W, $\dot{m}_a$, L_{EAHE}, and $D_{i,EAHE}$ and decreasing the D.
- An increment in the L, W, and $D_{i,EAHE}$ and a decrement in the D and L_{EAHE} are associated with a rise in the $\dot{E}$.
- As the $\dot{m}_a$ rises, the $\dot{E}$ initially boosts and then diminishes.
- In the optimal state ($L = 9.999$ m, $W = 5.0$ m, $D = 0.1$ m, $\dot{m}_a = 1.73$ kg/s, $D_{i,EAHE} = 0.5$ m, and $L_{EAHE} = 91.147$ m), the useful thermal and electrical power outputs of the system are $\dot{Q} = 159229.4$ Wh and $\dot{E} = 9242.2$ Wh, respectively.
- For the optimal HPT-EAHE system, the highest and lowest $\dot{E}$ respectively occur in July and December.
- During cold months, the optimal HPT-EAHE system can supply heating power ranging from 17701.4 to 25295.7 kWh, while in warm months, it can deliver cooling power between 3092.7 and 6738.9 kWh.
- The optimal HPT-EAHE unit is able to supply 26536.1 kWh of cooling load, 132693.3 kWh of heating load, and 9242.2 kWh of electrical load of the building per year.

CRediT authorship contribution statement

Amin Shahsavar: Writing – original draft, Supervision, Software, Data curation, Conceptualization. **Neda Azimi:** Writing – original draft, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

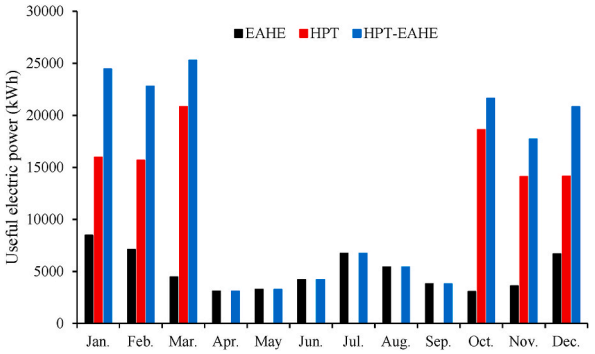


Fig. 10. The thermal performance of HPT, EAHE and HPT-EAHE systems throughout various months of the year.

Data availability

Data will be made available on request.

Appendix A. The value of decision-making variables and objective functions for the optimal points in the Pareto chart

	L (m)	W (m)	D (m)	$\dot{m}_a$ (kg/s)	$D_{i,EAHE}$ (m)	L_{EAHE} (m)	$\dot{Q}$ (Wh)	$\dot{E}$ (Wh)
1	9.999	5.000	0.100	1.730	0.500	91.147	9273.772	159211.288
2	9.998	5.000	0.101	1.709	0.500	91.147	9342.084	157879.537
3	10.000	4.999	0.100	1.056	0.500	90.997	10684.748	112558.064
4	10.000	5.000	0.100	0.857	0.500	90.939	10826.755	96633.702
5	9.998	4.999	0.101	1.682	0.500	91.141	9427.705	156141.476
6	9.999	5.000	0.100	1.029	0.499	91.092	10709.672	110494.838
7	10.000	5.000	0.100	0.898	0.500	91.001	10806.313	100028.433
8	9.999	5.000	0.100	1.702	0.500	91.144	9368.167	157398.289
9	9.999	5.000	0.100	0.992	0.500	91.024	10743.879	107605.582
10	9.999	5.000	0.100	0.878	0.500	90.979	10816.196	98379.733
11	9.998	5.000	0.100	1.372	0.500	91.116	10222.629	135540.573
12	9.999	5.000	0.100	1.383	0.500	91.095	10201.665	136264.739
13	9.999	5.000	0.100	1.642	0.500	91.018	9557.214	153495.671
14	9.999	5.000	0.100	1.500	0.500	91.071	9941.507	144181.695
15	9.998	5.000	0.101	1.655	0.500	91.143	9514.973	154357.471
16	10.000	5.000	0.100	0.911	0.500	90.965	10799.891	101072.778
17	9.999	5.000	0.100	1.567	0.500	91.116	9767.367	148641.861
18	9.999	5.000	0.100	0.930	0.500	90.983	10787.469	102629.737
19	9.998	5.000	0.100	1.286	0.500	91.113	10381.178	129520.708
20	9.999	5.000	0.100	1.590	0.500	91.113	9704.657	150140.616
21	9.999	5.000	0.100	1.532	0.500	91.115	9860.934	146313.324
22	9.999	5.000	0.100	1.509	0.500	91.147	9916.528	144845.497
23	9.999	5.000	0.100	1.196	0.500	91.036	10518.374	123006.480
24	9.999	5.000	0.100	1.051	0.500	91.073	10692.570	112184.664
25	9.999	5.000	0.100	1.064	0.500	91.080	10677.512	113213.836
26	9.999	5.000	0.100	1.696	0.500	91.145	9385.681	157018.347
27	9.999	5.000	0.100	0.863	0.500	90.959	10823.350	97129.682
28	9.999	5.000	0.100	1.542	0.500	91.075	9834.149	146942.572
29	10.000	5.000	0.100	1.346	0.500	91.035	10277.408	133674.487
30	9.998	5.000	0.100	1.248	0.500	91.004	10444.058	126747.173
31	10.000	5.000	0.100	1.133	0.500	91.024	10603.782	118431.193
32	9.999	4.999	0.100	1.212	0.500	91.061	10498.606	124168.140
33	10.000	5.000	0.100	0.869	0.500	90.948	10820.943	97634.381
34	9.998	5.000	0.100	1.663	0.500	91.140	9491.119	154915.363
35	9.999	5.000	0.100	0.959	0.500	91.006	10768.398	105029.832
36	9.999	5.000	0.100	1.421	0.500	91.026	10122.821	138811.465
37	9.999	5.000	0.100	1.182	0.500	91.017	10539.217	122028.087
38	9.999	5.000	0.100	1.090	0.500	91.069	10650.692	115188.662
39	10.000	5.000	0.100	1.001	0.500	90.982	10736.833	108338.416
40	9.999	5.000	0.100	1.405	0.500	91.029	10158.614	137740.843
41	9.999	5.000	0.100	0.968	0.500	90.951	10760.314	105689.463
42	9.999	5.000	0.100	1.543	0.500	91.099	9833.672	147036.163
43	9.999	5.000	0.100	1.356	0.500	91.046	10252.517	134373.846
44	9.999	5.000	0.100	1.141	0.500	91.071	10592.911	118975.477
45	9.998	5.000	0.100	1.669	0.500	91.140	9471.039	155313.430

(*continued on next page*)

(*continued*)

	L (m)	W (m)	D (m)	$\dot{m}_a$ (kg/s)	$D_{i,EAHE}$ (m)	L_{EAHE} (m)	$\dot{Q}$ (Wh)	$\dot{E}$ (Wh)
46	9.998	5.000	0.100	1.315	0.500	91.115	10330.517	131550.330
47	9.998	5.000	0.100	1.604	0.500	91.140	9665.701	151046.926
48	9.999	5.000	0.100	1.489	0.500	91.142	9968.359	143454.779
49	9.999	5.000	0.100	1.560	0.500	91.105	9787.930	148181.246
50	9.998	4.999	0.100	1.273	0.500	91.034	10401.515	128592.141
51	9.999	5.000	0.100	1.327	0.500	91.072	10311.088	132371.513
52	9.999	5.000	0.100	1.390	0.500	91.090	10188.631	136771.943
53	9.999	5.000	0.100	1.192	0.500	91.074	10526.366	122731.476
54	9.999	5.000	0.100	1.305	0.500	91.005	10350.142	130820.334
55	9.999	5.000	0.100	0.889	0.500	91.052	10809.286	99292.864
56	9.999	5.000	0.100	1.722	0.500	91.136	9301.349	158674.038
57	9.998	5.000	0.101	1.691	0.500	91.128	9398.794	156696.295
58	9.999	5.000	0.100	1.504	0.500	91.107	9929.991	144484.832
59	9.999	5.000	0.100	1.042	0.500	91.054	10700.307	111502.584
60	9.999	5.000	0.100	1.629	0.500	91.115	9594.098	152698.317
61	9.999	5.000	0.100	1.153	0.500	91.035	10576.892	119870.238
62	9.998	5.000	0.101	1.714	0.500	91.144	9328.024	158155.795
63	10.000	5.000	0.100	0.944	0.500	91.096	10777.976	103790.295
64	10.000	5.000	0.100	1.258	0.500	91.014	10428.901	127513.584
65	9.999	5.000	0.100	0.923	0.500	91.006	10791.250	102127.443
66	9.999	5.000	0.100	0.943	0.500	91.059	10778.954	103691.115
67	9.999	5.000	0.100	1.115	0.500	91.028	10624.727	117098.157
68	9.999	5.000	0.100	1.237	0.500	91.067	10459.412	126018.706
69	9.999	4.999	0.100	1.522	0.500	90.997	9884.679	145641.704
70	10.000	5.000	0.100	1.298	0.500	91.046	10363.425	130318.294

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